Worksheet — Efficiency and Energy Density Simulation

Use with *Simulation - Efficiency and energy density.html* · efficiency, Sankey diagrams and the energy density of fuels (Block 12)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/22-efficiency-energy-density/** in a browser (offline: *Simulation - Efficiency and energy density.html*). Choose the example at the top: **Bouncing ball · Motor lift · Sankey · Fuel tank**. Pick a preset in **Example** or set the values yourself (type a value and press Enter).

Press ▶️ Start (not needed for the Sankey builder); ◄ / ► step through the motion. The Sankey builder draws the diagram to scale on squared paper; one output is found by conservation.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 20 min) · **Part 3** homework (≈ 30 min). efficiency = useful output ÷ total input · E = energy density × mass · 1 kWh = 3.6×10^6 J. g = 9.81 N kg^{-1}. Answers to 3 s.f.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. 🏀 A tennis ball, a bouncy ball and a plasticine ball are dropped from 1.0 m. Which keeps the most energy in the bounce? Which the least?			
2. A tennis ball keeps 55 % of its energy in each bounce. What fraction of its starting energy does it have after two bounces?			
3. Spot the mistake: 'the motor's efficiency is 120 %'.			
4. Spot the mistake: a 10 % sound arrow is drawn wider than the 60 % thermal arrow.			
5. ⛽ Why is an electric car's battery so much heavier than a petrol car's full tank?			

6. A tennis ball dropped from 1.00 m bounces back to 55 cm. What fraction of its energy did it keep? Check: Bouncing ball → Class demo: tennis ball.

Working

7. (retrieval) Convert 1 kWh to joules.

Working

8. (slide 13) An electric motor takes in 2.0 kW. Its useful output is 1.4 kW and 0.5 kW is thermal; the rest is sound. Find the sound output and the efficiency. Check: Sankey → Slide 13.

Working

9. (exit question) Petrol stores 46 MJ kg^{-1}. What mass of petrol stores 920 MJ?

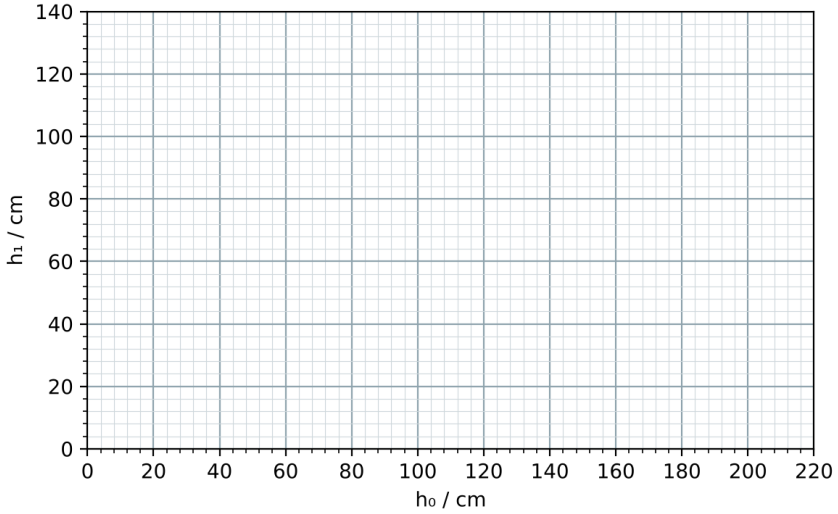
Working

Part 2 — During the lab

Bouncing ball demo. Drop each ball from 1.0 m next to a metre stick and film it; read the first bounce height h_1. Simulation: run each class-demo ball.

ball	h_0 / cm	h_1 (class) / cm	h_1 (simulation) / cm	energy kept = h_1/h_0	h_2 (simulation) / cm
tennis ball	100				
bouncy ball	100				
plasticine	100				

10. Drop the tennis ball from 50, 80, 100, 150 and 200 cm (● Record each). Plot h_1 against h_0 and find the gradient. What does it tell you?

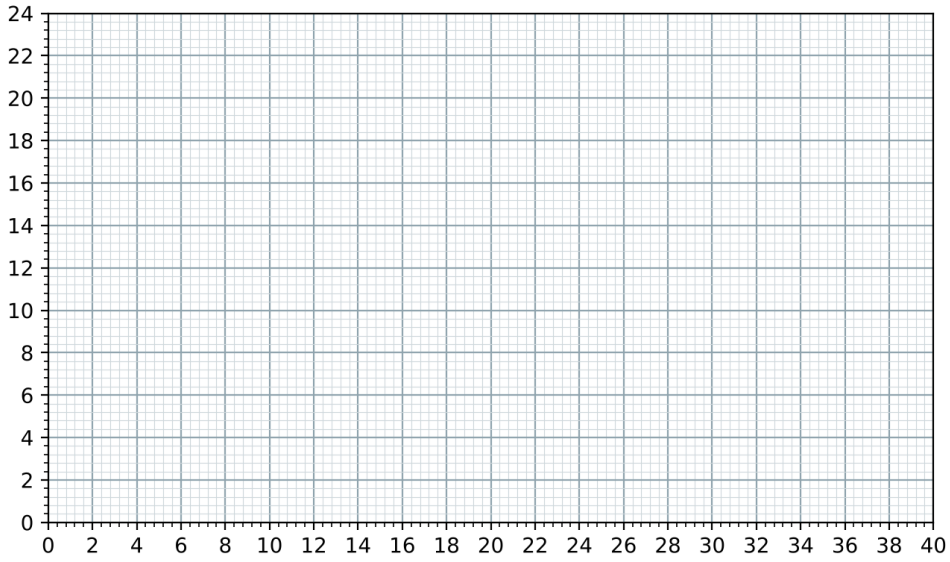


Gradient = _____ .

11. (Oxford worked example) A motor raises a 150 N weight through 7.2 m. The energy supplied to it is 3.5×10^3 J. Calculate the useful work and the efficiency. Check: Motor lift → Oxford preset.

Working

12. Draw the slide 13 motor's Sankey diagram to scale on the squared paper (1 square = 100 W). Read the efficiency from the widths.



Part 3 — Homework

13. (exit question) A motor takes in 500 W and gives 350 W of useful power. Calculate its efficiency and sketch the Sankey diagram.

Working

14. (slide 15) The 2.0 kW motor of question 8 runs for one hour. Calculate the energy wasted.

Working

15. (slide 18) A car carries 40 kg of petrol (46 MJ kg^{-1}). Its engine is 25 % efficient and it needs 20 kW of useful power at 25 m s^{-1}. Calculate the energy stored, the useful energy, the time and the distance on one tank. Check: Fuel tank → Slide 18.

Working

16. An electric car carries 400 kg of Li-ion cells (0.9 MJ kg^{-1}, 90 % efficient) and needs the same 20 kW at 25 m s^{-1}. How far does it go? Why are batteries heavy?

Working

17. What mass of uranium (≈ 8 × 10^{7} MJ kg^{-1}) releases the same energy as 40 kg of petrol? Give one reason why cars do not use it.

Working

18. (slide 16: spot the mistake) A: input 100 J → useful 60 J, thermal 30 J. B: 'efficiency = 75 J wasted ÷ 100 J = 75 %'. What is wrong with each?